

	$\mathcal{K}_{0+}^{\#1}$	$\mathcal{K}_{0-}^{\#1} \alpha\beta\chi$				
$\mathcal{K}_{0+}^{\#1} \dagger$	0	0				
$\mathcal{K}_{0-}^{\#1} \dagger^{\alpha\beta\chi}$	0	0	$\mathcal{K}_{1+}^{\#1} \alpha\beta$	$\mathcal{K}_{1+}^{\#2} \alpha\beta$	$\mathcal{K}_{1-}^{\#1} \alpha$	$\mathcal{K}_{1-}^{\#2} \alpha$
	$\mathcal{K}_{1+}^{\#1} \dagger^{\alpha\beta}$	$-\frac{\gamma_1 k^2}{2}$	$\frac{k^2 \binom{4}{\kappa_4}}{2\sqrt{2}}$	0	0	
	$\mathcal{K}_{1+}^{\#2} \dagger^{\alpha\beta}$	$\frac{k^2 \binom{4}{\kappa_4}}{2\sqrt{2}}$	$\frac{\gamma_2 k^2}{2}$	0	0	
	$\mathcal{K}_{1-}^{\#1} \dagger^{\alpha}$	0	0	$-k^2 \binom{4}{\kappa_2}$	$\frac{k^2 \binom{4}{\kappa_2}}{\sqrt{2}}$	
	$\mathcal{K}_{1-}^{\#2} \dagger^{\alpha}$	0	0	$\frac{k^2 \binom{4}{\kappa_2}}{\sqrt{2}}$	$-\frac{k^2 \binom{4}{\kappa_2}}{2}$	$\mathcal{K}_{2+}^{\#1} \alpha\beta$ $\mathcal{K}_{2-}^{\#1} \alpha\beta\chi$
				$\mathcal{K}_{2+}^{\#1} \dagger^{\alpha\beta}$	$\frac{3\gamma_3 k^2}{2}$	0
				$\mathcal{K}_{2-}^{\#1} \dagger^{\alpha\beta\chi}$	0	0

[illegible]

Abbreviations used in matrices	
$Y_1 = 3 \left(\overset{(4)}{\kappa_1} + \overset{(4)}{\kappa_2} \right) + 2 \overset{(4)}{\kappa_3}$	$\&\& Y_2 = \overset{(4)}{\kappa_3} + \overset{(4)}{\kappa_4}$
$\&\& Y_3 = \overset{(4)}{\kappa_1} + \overset{(4)}{\kappa_2}$	$\&\&$
$\text{Det}(1^+) = -\frac{1}{8} k^4 \left(6 \overset{(4)}{\kappa_1} \left(\overset{(4)}{\kappa_3} + \overset{(4)}{\kappa_4} \right) + 6 \overset{(4)}{\kappa_2} \left(\overset{(4)}{\kappa_3} + \overset{(4)}{\kappa_4} \right) + \left(2 \overset{(4)}{\kappa_3} + \overset{(4)}{\kappa_4} \right)^2 \right)$	
$\&\& \text{Det}(2^+) = \frac{3}{2} k^2 \left(\overset{(4)}{\kappa_1} + \overset{(4)}{\kappa_2} \right)$	

$$\begin{aligned} & \text{Lagrangian} \\ & \kappa_1^{(4)} \partial_\beta \mathcal{K}^\delta_{\chi\delta} \partial^\chi \mathcal{K}^\alpha_{\alpha\beta} + \kappa_2^{(4)} \partial_\chi \mathcal{K}^\delta_{\beta\delta} \partial^\chi \mathcal{K}^\alpha_{\alpha\beta} + \kappa_3^{(4)} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\alpha\chi} + \kappa_4^{(4)} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} - 3 \kappa_1^{(4)} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\chi\alpha} - \\ & 3 \kappa_2^{(4)} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\chi\alpha} - \kappa_3^{(4)} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\chi\alpha} + \frac{1}{2} \kappa_3^{(4)} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} + \frac{1}{2} \kappa_4^{(4)} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} \end{aligned}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{T}_{\alpha\beta\chi}$	
Source constraint(s)	# constraint(s)	Covariant form
$\mathcal{T}_0^{\#1\alpha\beta\chi} == 0$	1	$\partial_\delta \partial^\alpha \mathcal{T}^{\beta\chi\delta} + \partial_\delta \partial^\alpha \mathcal{T}^{\delta\beta\chi} + \partial_\delta \partial^\beta \mathcal{T}^{\chi\alpha\delta} + \partial_\delta \partial^\chi \mathcal{T}^{\alpha\beta\delta} + \partial_\delta \partial^\chi \mathcal{T}^{\delta\alpha\beta} + \partial_\delta \partial^\delta \mathcal{T}^{\beta\alpha\chi} ==$ $\partial_\delta \partial^\alpha \mathcal{T}^{\chi\beta\delta} + \partial_\delta \partial^\beta \mathcal{T}^{\alpha\chi\delta} + \partial_\delta \partial^\beta \mathcal{T}^{\delta\alpha\chi} + \partial_\delta \partial^\chi \mathcal{T}^{\beta\alpha\delta} + \partial_\delta \partial^\delta \mathcal{T}^{\alpha\beta\chi} + \partial_\delta \partial^\delta \mathcal{T}^{\chi\alpha\beta}$
$\mathcal{T}_0^{\#1} == 0$	1	$\partial_\beta \mathcal{T}^\alpha{}_\alpha{}^\beta == 0$
$\mathcal{T}_1^{\#1\alpha} + 2 \mathcal{T}_1^{\#2\alpha} == 0$	3	$\partial_\chi \partial^\alpha \mathcal{T}^\beta{}_\beta{}^\chi + \partial_\chi \partial^\chi \mathcal{T}^{\beta\alpha}{}_\beta == 3 \partial_\chi \partial_\beta \mathcal{T}^{\beta\alpha\chi}$
$\mathcal{T}_2^{\#1\alpha\beta\chi} == 0$	5	$3 \partial_\epsilon \partial_\delta \partial^\chi \partial^\alpha \mathcal{T}^{\delta\beta\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial^\chi \partial^\alpha \mathcal{T}^{\delta\beta}{}_\delta + 2 \partial_\epsilon \partial^\delta \partial_\delta \partial^\beta \mathcal{T}^{\alpha\chi\delta} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{T}^{\chi\alpha\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{T}^{\delta\alpha\chi} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{T}^{\beta\alpha\delta} +$ $4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{T}^{\delta\alpha\beta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{T}^{\alpha\beta\chi} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\alpha \mathcal{T}^{\delta}{}_\delta{}^\epsilon + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{T}^{\delta\beta\epsilon} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\epsilon \mathcal{T}^{\delta\alpha}{}_\delta ==$ $3 \partial_\epsilon \partial_\delta \partial^\chi \partial^\beta \mathcal{T}^{\delta\alpha\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial^\chi \partial^\beta \mathcal{T}^{\delta\alpha}{}_\delta + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{T}^{\beta\chi\delta} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{T}^{\chi\beta\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{T}^{\delta\beta\chi} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{T}^{\alpha\beta\delta} +$ $2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{T}^{\beta\alpha\chi} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{T}^{\chi\alpha\beta} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\beta \mathcal{T}^{\delta}{}_\delta{}^\epsilon + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{T}^{\delta\alpha\epsilon} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\epsilon \mathcal{T}^{\delta\beta}{}_\delta$

Total # constraint(s):	10		
Resolved pole(s)	# polarization(s)	Square mass	Residue

	2	0	$ \begin{aligned} & (54 \kappa_2^{(4)3} + 54 \kappa_2^{(4)2} \kappa_3^{(4)} + 20 \kappa_2^{(4)} \kappa_3^{(4)2} + 36 \kappa_2^{(4)2} \kappa_4^{(4)} + 20 \kappa_2^{(4)} \kappa_3^{(4)} \kappa_4^{(4)} + 5 \kappa_2^{(4)} \kappa_4^{(4)2} + \\ & 18 \kappa_1^{(4)2} (3 \kappa_2^{(4)} + \kappa_3^{(4)} + \kappa_4^{(4)}) + 3 \kappa_1^{(4)} (36 \kappa_2^{(4)2} + (2 \kappa_3^{(4)} + \kappa_4^{(4)})^2 + 6 \kappa_2^{(4)} (4 \kappa_3^{(4)} + 3 \kappa_4^{(4)})) - \\ & \sqrt{(324 \kappa_1^{(4)4} (9 \kappa_2^{(4)2} - 2 \kappa_2^{(4)} (\kappa_3^{(4)} + \kappa_4^{(4)}) + (\kappa_3^{(4)} + \kappa_4^{(4)})^2) + \\ & \kappa_2^{(4)2} (2916 \kappa_2^{(4)4} + 7 (2 \kappa_3^{(4)} + \kappa_4^{(4)})^4 + 648 \kappa_2^{(4)3} (5 \kappa_3^{(4)} + 2 \kappa_4^{(4)}) + 36 \kappa_2^{(4)} (2 \kappa_3^{(4)} + \kappa_4^{(4)})^2 (4 \kappa_3^{(4)} + 3 \kappa_4^{(4)}) + \\ & 108 \kappa_2^{(4)2} (15 \kappa_3^{(4)2} + 16 \kappa_3^{(4)} \kappa_4^{(4)} + 5 \kappa_4^{(4)2})) + 9 \kappa_1^{(4)2} (1944 \kappa_2^{(4)4} + (2 \kappa_3^{(4)} + \kappa_4^{(4)})^4 + \\ & 216 \kappa_2^{(4)3} (4 \kappa_3^{(4)} + \kappa_4^{(4)}) + 4 \kappa_2^{(4)} (2 \kappa_3^{(4)} + \kappa_4^{(4)})^2 (6 \kappa_3^{(4)} + 7 \kappa_4^{(4)}) + 24 \kappa_2^{(4)2} (7 \kappa_3^{(4)2} + 13 \kappa_3^{(4)} \kappa_4^{(4)} + 7 \kappa_4^{(4)2})) + \\ & 108 \kappa_1^{(4)3} (108 \kappa_2^{(4)3} + 6 \kappa_2^{(4)2} (2 \kappa_3^{(4)} - \kappa_4^{(4)}) + (\kappa_3^{(4)} + \kappa_4^{(4)}) (2 \kappa_3^{(4)} + \kappa_4^{(4)})^2 + \kappa_2^{(4)} (4 \kappa_3^{(4)2} + 14 \kappa_3^{(4)} \kappa_4^{(4)} + 9 \kappa_4^{(4)2})) + \\ & 12 \kappa_1^{(4)} \kappa_2^{(4)} (972 \kappa_2^{(4)4} + 21 \kappa_2^{(4)} (\kappa_3^{(4)} + \kappa_4^{(4)}) (2 \kappa_3^{(4)} + \kappa_4^{(4)})^2 + (2 \kappa_3^{(4)} + \kappa_4^{(4)})^4 + \\ & 54 \kappa_2^{(4)3} (14 \kappa_3^{(4)} + 5 \kappa_4^{(4)}) + 9 \kappa_2^{(4)2} (28 \kappa_3^{(4)2} + 34 \kappa_3^{(4)} \kappa_4^{(4)} + 13 \kappa_4^{(4)2}))) / \\ & (\kappa_2^{(4)} (\kappa_1^{(4)} + \kappa_2^{(4)}) (6 \kappa_1^{(4)} (\kappa_3^{(4)} + \kappa_4^{(4)}) + 6 \kappa_2^{(4)} (\kappa_3^{(4)} + \kappa_4^{(4)}) + (2 \kappa_3^{(4)} + \kappa_4^{(4)})^2)) \end{aligned} $
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	2	0	$ \begin{aligned} & (54 \binom{(4)}{k_2^2} + 54 \binom{(4)}{k_2^2} \binom{(4)}{k_3} + 20 \binom{(4)}{k_2} \binom{(4)}{k_3^2} + 36 \binom{(4)}{k_2^2} \binom{(4)}{k_4} + 20 \binom{(4)}{k_2} \binom{(4)}{k_3} \binom{(4)}{k_4} + 5 \binom{(4)}{k_2} \binom{(4)}{k_4^2} + \\ & 18 \binom{(4)}{k_1^2} (3 \binom{(4)}{k_2} + \binom{(4)}{k_3} + \binom{(4)}{k_4}) + 3 \binom{(4)}{k_1} (36 \binom{(4)}{k_2^2} + (2 \binom{(4)}{k_3} + \binom{(4)}{k_4})^2 + 6 \binom{(4)}{k_2} (4 \binom{(4)}{k_3} + 3 \binom{(4)}{k_4})) + \\ & \sqrt{(324 \binom{(4)}{k_1^4} (9 \binom{(4)}{k_2^2} - 2 \binom{(4)}{k_2} (\binom{(4)}{k_3} + \binom{(4)}{k_4}) + (\binom{(4)}{k_3} + \binom{(4)}{k_4})^2) + \\ & \binom{(4)}{k_2^2} (2916 \binom{(4)}{k_2^4} + 7 (2 \binom{(4)}{k_3} + \binom{(4)}{k_4})^4 + 648 \binom{(4)}{k_2^3} (5 \binom{(4)}{k_3} + 2 \binom{(4)}{k_4}) + 36 \binom{(4)}{k_2} (2 \binom{(4)}{k_3} + \binom{(4)}{k_4})^2 (4 \binom{(4)}{k_3} + 3 \binom{(4)}{k_4}) + \\ & 108 \binom{(4)}{k_2^2} (15 \binom{(4)}{k_3^2} + 16 \binom{(4)}{k_3} \binom{(4)}{k_4} + 5 \binom{(4)}{k_4^2})) + 9 \binom{(4)}{k_1^2} (1944 \binom{(4)}{k_2^4} + (2 \binom{(4)}{k_3} + \binom{(4)}{k_4})^4 + \\ & 216 \binom{(4)}{k_2^3} (4 \binom{(4)}{k_3} + \binom{(4)}{k_4}) + 4 \binom{(4)}{k_2} (2 \binom{(4)}{k_3} + \binom{(4)}{k_4})^2 (6 \binom{(4)}{k_3} + 7 \binom{(4)}{k_4}) + 24 \binom{(4)}{k_2^2} (7 \binom{(4)}{k_3^2} + 13 \binom{(4)}{k_3} \binom{(4)}{k_4} + 7 \binom{(4)}{k_4^2})) + \\ & 108 \binom{(4)}{k_1^3} (108 \binom{(4)}{k_2^3} + 6 \binom{(4)}{k_2^2} (2 \binom{(4)}{k_3} - \binom{(4)}{k_4}) + (\binom{(4)}{k_3} + \binom{(4)}{k_4}) (2 \binom{(4)}{k_3} + \binom{(4)}{k_4})^2 + \binom{(4)}{k_2} (4 \binom{(4)}{k_3^2} + 14 \binom{(4)}{k_3} \binom{(4)}{k_4} + 9 \binom{(4)}{k_4^2})) + \\ & 12 \binom{(4)}{k_1} \binom{(4)}{k_2} (972 \binom{(4)}{k_2^4} + 21 \binom{(4)}{k_2} (\binom{(4)}{k_3} + \binom{(4)}{k_4}) (2 \binom{(4)}{k_3} + \binom{(4)}{k_4})^2 + (2 \binom{(4)}{k_3} + \binom{(4)}{k_4})^4 + \\ & 54 \binom{(4)}{k_2^3} (14 \binom{(4)}{k_3} + 5 \binom{(4)}{k_4}) + 9 \binom{(4)}{k_2^2} (28 \binom{(4)}{k_3^2} + 34 \binom{(4)}{k_3} \binom{(4)}{k_4} + 13 \binom{(4)}{k_4^2}))) / \\ & (\binom{(4)}{k_2} (\binom{(4)}{k_1} + \binom{(4)}{k_2}) (6 \binom{(4)}{k_1} (\binom{(4)}{k_3} + \binom{(4)}{k_4}) + 6 \binom{(4)}{k_2} (\binom{(4)}{k_3} + \binom{(4)}{k_4}) + (2 \binom{(4)}{k_3} + \binom{(4)}{k_4})^2)) \end{aligned} $
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[illegible]